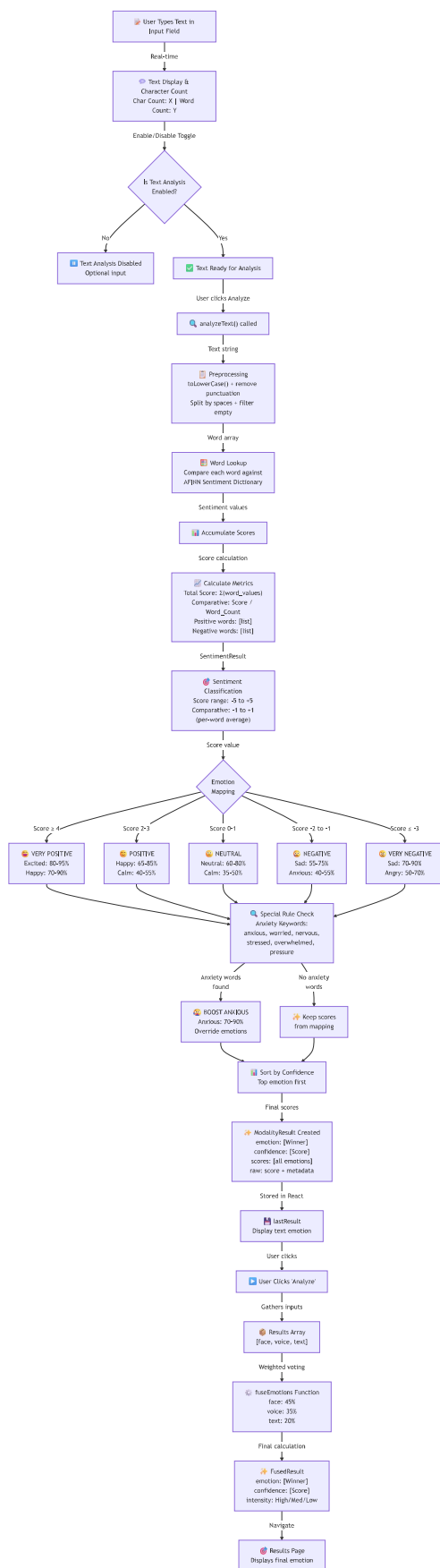






AFINN Sentiment Dictionary

Lightweight, vocabulary-based approach (no ML models needed)

Dictionary of ~400 words with scores:

Positive: 1 to +3

Negative: -3 to -1

Example scoring:

"amazing" → +3

"great" → +3

"good" → +2

"hate" → -3

"bad" → -2

"tired" → -1



Analysis Process

1. **Clean text:** Remove punctuation, convert to lowercase
2. **Tokenize:** Split into words
3. **Look up:** Match each word in AFINN dictionary
4. **Accumulate:** Sum all word sentiment scores
5. **Calculate:** Total score and comparative (average)
6. **Extract:** Positive/negative word lists



Sentiment Score → Emotion Mapping

Score ≥ 4 **VERY POSITIVE / EXCITED**

Example: "I absolutely love this! It's amazing and wonderful!"

Confidence: Excited 80-95%, Happy 70-90%

Score 2-3 **POSITIVE / HAPPY**

Example: "This is great and I'm happy about it."

Confidence: Happy 65-85%, Calm 40-55%

Score 0-1 **NEUTRAL**

Example: "It's okay. Nothing special, just fine."

Confidence: Neutral 60-80%, Calm 35-50%

Score -2 to -1 **NEGATIVE / SAD**

Example: "I'm sad and feeling bad about this."

Confidence: Sad 55-75%, Anxious 40-55%

Score ≤ -3 **VERY NEGATIVE / ANGRY**

Example: "I hate this! It's terrible and awful!"

Confidence: Sad 70-90%, Angry 50-70%, Fearful 35-50%

 Common AFINN Words Examples

happy (+3)

love (+3)

great (+3)

amazing (+3)

excited (+3)

wonderful (+3)

good (+2)

nice (+2)

glad (+2)

enjoy (+2)

sad (-2)

bad (-2)

hate (-3)

angry (-3)

afraid (-2)

worried (-2)

nervous (-2)

stressed (-2)

tired (-1)

bored (-1)



Real Example Analysis

Input Text: "I'm feeling happy and excited! This is amazing!"

Words: [I'm, feeling, happy, and, excited, This, is, amazing]

AFINN Lookup:

- "i'm" → 0 (not in dictionary)
- "feeling" → 0
- "happy" → +3 ✓ (positive)
- "and" → 0
- "excited" → +3 ✓ (positive)
- "this" → 0
- "is" → 0
- "amazing" → +3 ✓ (positive)

Total Score: 3 + 3 + 3 = 9 (capped at ~4)

Comparative: 9 / 8 ≈ 1.1

Positive Words: [happy, excited, amazing]

Negative Words: []

Classification: Score ≥ 4 → VERY POSITIVE

Result:

- Emotion: Excited (95% confidence)
- Scores: [Excited: 95, Happy: 80]

⚡ Special Rule: Anxiety Boost

If any anxiety-related keywords are detected, the emotion is overridden to ANXIOUS:

Anxiety Keywords:

anxious, worried, nervous,
stressed, overwhelmed, pressure

Example: "I'm worried about tomorrow"

- Normal score: -2 (negative)
- Would map to: Sad
- But "worried" detected → BOOST ANXIOUS
- Final: Anxious (70-90%)